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Video Frame Interpolation Approach for Reservoir Simulation Visualization Utilizing Stable Diffusion Model

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Abstract

Reservoir simulation is a fundamental yet computationally intensive task in oil and gas exploration and production. It involves solving complex mathematical equations over time to predict the behavior of subsurface fluid flow under varying geological and operational conditions. This task is essential for evaluating production strategies, optimizing field development, and making informed operational decisions. To operate these complex systems, engineers leverage high-performance computers to execute such simulations at high spatial and temporal resolutions to handle the substantial computational power required. Moreover, this task also demands high storage capacity to store 3D reservoir models, which comprise a large number of grids representing the reservoir's geological structure, as well as the dynamic properties of the fluid, including water and oil saturation, pressure, and temperature, among others. In this work, we present a novel approach that utilizes the concept of video frame interpolation, a well-known problem in the computer vision field, to enhance reservoir simulation capabilities by increasing its resolution without affecting computational or storage capacity. The objective of this study is to improve simulation results with minimal computational processing. Our proposed approach employs a type of Generative AI Model, specifically a Diffusion Model, which utilizes a U-net architecture as its backbone to train the model. The results of our framework demonstrate remarkable alignment with outputs from full-scale simulation runs, achieving visually and numerically consistent results with only minimal deviations.

Keywords: Reservoir Simulation, Visualization, Generative AI, Diffusion Model, Video Frame Interpolation, In-betweening

Introduction

In today's numerous critical applications, simulation technologies are pivotal in predicting outcomes and evaluating events using mathematical models over time. Developing these systems requires careful configuration of initial inputs, conditions, and parameters. A visualization tool is necessary to analyze the simulation results effectively, as it enables studying how dynamic properties respond to changes in user input. However, operating these systems and visualizing their results comes at a significant cost, as they demand substantial computational resources, resulting in considerable processing time and data storage (Luan, 2024). In the oil and gas industry, reservoir simulation is vital to observe the behavior of the dynamic fluids during production. Currently, reservoir engineers face considerable difficulties in visualizing these large-scale models. This work addresses the significant technical challenges of visualizing large-scale reservoir simulation models, which can comprise billions of cells. A considerable bottleneck lies in rendering 3D dynamic properties (e.g., saturation and pressure maps) over time, which requires enormous computational resources and storage capacity. To illustrate, storing a single property of a 1-billion-cell model at float-precision 32 (fp32) would necessitate approximately 29 gigabytes per timestep, resulting in over a terabyte of data for merely 40 timesteps. Furthermore, effectively visualizing these massive datasets demands advanced visualization techniques, given that even high-performance graphics processing units (GPUs) cannot simultaneously handle such vast amounts of data. As a temporary solution, engineers often output grid data at extremely low frequencies, consequently overlooking numerous valuable insights.

Recently, we have seen many contributions by engineers and geoscientists to adopt machine learning frameworks in reservoir modeling. For instance, the work of (Doloi, Ghosh, & Phirani, 2023) utilizes Generative Adversarial Networks GANs to predict the fine-scale fluid flow behavior out of a low-resolution of upscaled reservoir model. The authors proposed a reconstruction approach to produce a high-resolution model out of the low-resolution one. The goal is to use an upscaled reservoir model to conduct the simulation; after that, the upscaled model will be projected to a low-scale model to capture the fine-scale details of the simulation with minimum computational resources. Similarly, (Zhong, Y. Sun, Wang, & Ren, 2020) developed a proxy model using a conditional deep convolutional generative adversarial network (cDC-GAN), which enables rapid prediction of dynamic properties under waterflooding conditions.

In this work, we leveraged the current advancements in computer vision technology to address this challenge from a visualization viewpoint. Our methodology relies on the concept of video frame interpolation, also known as in-betweening, which aims to generate intermediate video frames that are conditioned on given pre- and post-frames (Chen, et al., 2025). This concept has many applications across different fields, including cartoon animation, enhancing video resolution, and compression (Tulyakov, et al., 2021). The goal of this work is to improve the simulator's capability with minimum computation interference by transforming the complex shape of a 3D reservoir model into 2D slices. By stacking these images, we can generate video samples that enable us to deal with this challenge as a video generation problem, reducing the complexity by a significant magnitude.

The first section of this paper discusses the SPE-10 dataset and the preprocessing that has been conducted to prepare the data and generate video samples to be used to train our machine learning model. In the second section, we introduce our proposed framework that relies on a diffusion model. In this part, we will go into the details of the implementation of the architecture and data partitioning. Following that, we showcase the results of the training process and the blind testing results, which were assessed by special performance metrics for the video generation task. Finally, we conclude by representing the limitations of this approach and the way forward to address these limitations to be adopted for a real-life complex reservoir model.

Dataset and Data Preprocessing

For this work, we benchmarked our framework using a public dataset of the SPE-10 models, which consists of 20 simulation models. Each model is a grid of 85 (layers)* 220 (height)* 60 (width), which includes the reservoir properties across 61 timesteps.

At this stage, a script was built to generate 5100 video samples for water saturation property, each frame is 220*60 in size. We first split the 85 layers, then sliced each layer into three portions across the time frame for the models, which is 61 timesteps. Each portion consists of 10 sliced images (frames), and we stack these images to form a video. So, three videos were generated from a single layer, each with 10 frames. Figure 1 illustrates the generation process of the video samples out of the 3D SPE-10 Models.

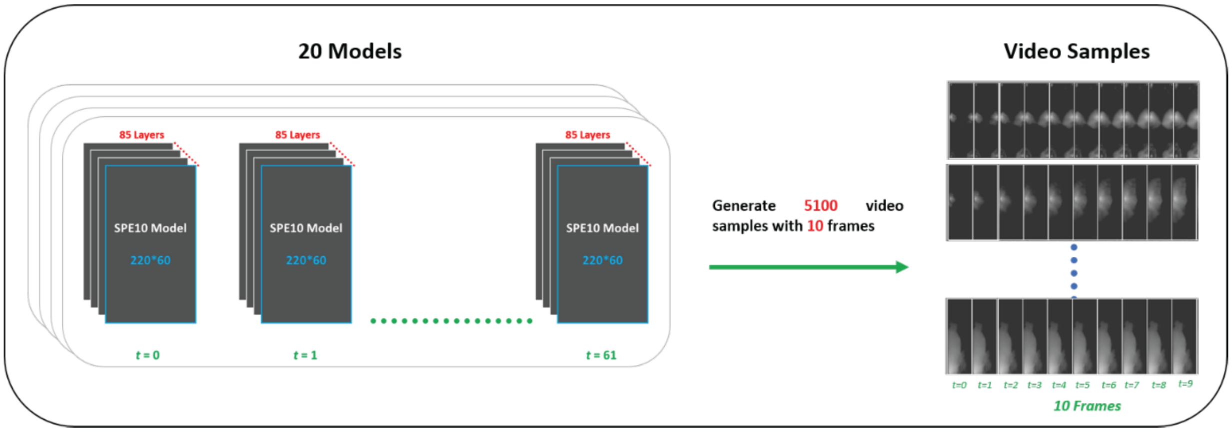

Figure 1—Generating video samples, each with 10 frames

In addition to water saturation, this process can also be repeated to generate video samples out of other dynamic properties such as pressure, temperature, and oil saturation, but for simplicity and proof of concept, this work focuses only on the generation of water saturation property.

Method

After the generation of the video samples, we split the data into 60% for training, 20% for testing, and validation. We chose Diffusion Models for the modeling component, as they have been extensively used in the literature to solve video generation problems. A Diffusion model is basically a type of deep generative model that operates in two main phases: a forward diffusion phase and a reverse diffusion phase. The original data is progressively corrupted during the forward phase by adding Gaussian noise over multiple steps. In the reverse phase, the model learns to reconstruct the original data by gradually denoising it, effectively reversing the diffusion process one step at a time (Croitoru, Hondru, Ionescu, & Shah, 2023).

Our framework is based on the work of (Voleti, Jolicoeur-Martineau, & Pal, 2022), which designed a methodology using the Diffusion Model for different video synthesis tasks, including video prediction, generation, and Interpolation. In this work, we will focus on the task of video interpolation, which aims to generate non-existing frames between two consecutive frames in a video sample (Dong, Ota, & Dong, 2023). Figure 2 illustrates the methodology of this work. The framework offers a novel solution to address the Visualization challenge posed by massive reservoir simulation models. Our approach leverages Diffusion Models to perform interpolation-based visualization of 3D data from 2D views, conditioned on the available low-frequency outputted grid data. This methodology enables efficient generation of intermediate states between sparse timepoints, with minimal decrease in accuracy, thereby circumventing the need for excessive computational resources.

During the visualization process, our framework operates as follows: first, a fixed viewpoint of the 3D model is established, followed by selecting two distant timeframes (typically 4-5 years apart). These distant timepoints are then used to generate interpolated intermediate stages via the Diffusion Model. Subsequently, the generated intermediates are re-projected onto the current 3D viewpoint, representing fluid dynamics. In the following section, we thoroughly explain the implementation and the training process. Moreover, our method can be adapted for seamless generalization across various levels of detail by incorporating additional variables and conditions, such as model resolution, property types, and others.

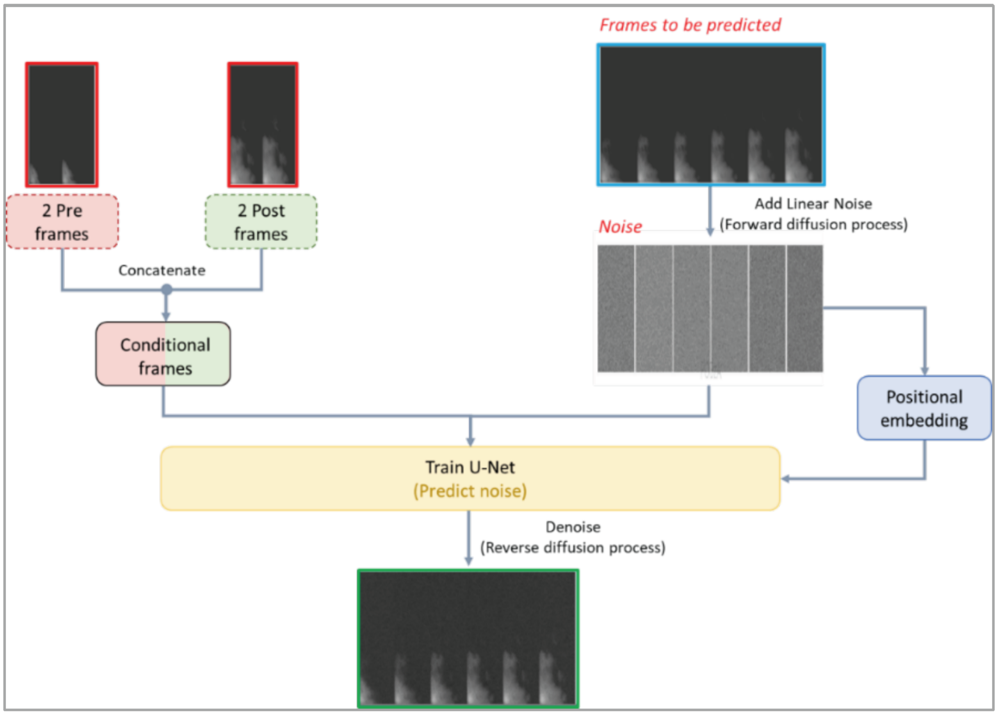

Figure 2—Proposed Methodology

Results and Discussion

The dataset utilized in this study was systematically divided to support the model development and evaluation pipeline. Specifically, we allocated 12 distinct models for training the core framework, while reserving four models each for validation and blind testing purposes. As introduced previously, this framework is based on the Stable Diffusion Model. The model is designed to process temporal information by taking *n* preceding frames and *n* succeeding frames, thereby leveraging this temporal information to perform high-quality video frame interpolation. The primary task of the model is to synthesize the intermediate frames that are conditioned on the given temporal context.

The implementation of the framework includes two stages. The first stage is to transform the frames that need to be predicted to complete random noise; we refer to this step as the **Forward Diffusion Process (FDP)**. While the second stage includes training a U- net network on these frames to predict the noise in each step and iteratively subtract (sample) the predicted noise from the noisy frames until it reaches the original frame, we refer to this step as the **Reverse Diffusion Process (RDP)**. To aid the model in capturing the progression of the diffusion process, we integrate **positional embeddings** that encode the noise level associated with each timestep. These embeddings allow the model to contextualize its predictions within the denoising sequence. This training process goes for each frame in each video sample. Figure 2 illustrates the model architecture and the training process.

During the model training, the forward process is executed as a fixed, non-trainable pipeline that applies increasing levels of Gaussian noise over time. On the other hand, the reverse process is learned by optimizing the U-Net to minimize the discrepancy between the predicted and the true noise. For this learning objective, we adopt the **Denoising Diffusion Probabilistic Model (DDPM)** sampling approach to control the denoising process. For the purpose of this study, we focused on **water saturation (SWAT)** as the target property for video generation and interpolation. However, depending on the desired application, the underlying framework is inherently flexible and can be extended to model other reservoir variables such as oil saturation, pressure, or temperature.

Initially, our training experiments were conducted using more extended video sequences comprising 19 frames in total. The model was conditioned on four frames—two prior and two post—and trained to generate the remaining 15 intermediate frames. However, after approximately 500,000 training steps, the outputs exhibited significant residual noise, indicating that the model had not yet converged to a satisfactory solution. To mitigate this, we restructured the training protocol by reducing the video sequence length to 10 frames, wherein four frames served as conditional input and six were designated as targets for generation. This adjustment allowed for more manageable training dynamics, faster convergence, and a more precise evaluation of the model's generative capacity under constrained conditions. Subsequently, the network was trained on two prior and two post frames, based on the specific property as a conditional frame to guide the training process to predict six frames in between. Our approach was assessed using **Frechet Video Distance (FVD),** which was proposed by (Unterthiner, et al., 2019), **Peak Signal to Noise Ratio (PSNR),** and **Structural Similarity Index Measure (SSIM)** metrics across a 500,000-step training where six frames were predicted given four conditioned inputs. These results are summarized in Table 1 below.

Table 1—Training Performance

Metric	FVD	PSNR	SSIM
Training on 3060 video samples	723	28.2	0.34

Following the training phase, we have conducted a blind test to assess the model's performance on unseen data. And for this task, we utilized the four models of the SPE-10 dataset to test 1020 video samples. At first, we started to feed the trained framework with pure noisy frames to be interpolated, as well as the four conditioned frames, the priors, and the posts. As a result, our trained framework could generalize to unseen data and predicted the frames in-between with FVD = 657, PSNR = 28.5, and SSIM = 0.36. Figure 3 shows the results of a few samples of the testing stage.

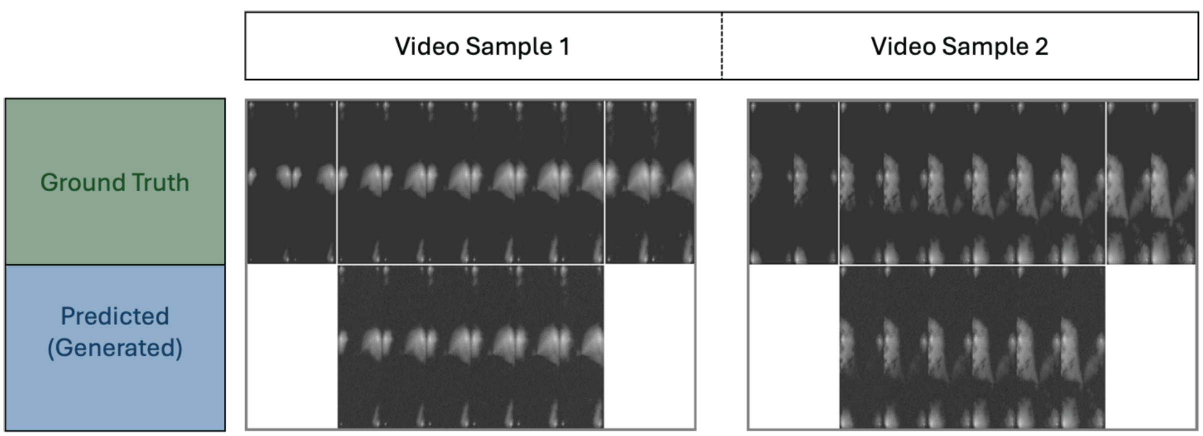

Figure 3—Blind testing results on two video samples: The bottom frames are the frames generated by the model, and the above are the ground truth generated by the simulator.

Conclusion

In conclusion, this work builds upon the foundational research that introduced a framework for both conditional and unconditional video frame generation. We adapt this concept to the domain of reservoir simulation visualization, with a particular focus on conditional video generation techniques that interpolate intermediate frames between multiple given images. Our contribution extends this idea further by exploring how these techniques can be applied to reconstruct three-dimensional views of reservoir models using temporally aligned 2D slices. Unlike prior approaches, which primarily utilize generative AI to create proxy models or upscale coarse-resolution simulations, our method focuses on enhancing the resolution and interpretability of 2D reservoir simulation outputs by applying Stable Diffusion Models. In our current implementation, the framework was trained to generate six intermediate frames conditioned on four input frames. This enables the reconstruction of continuous visual transitions within simulation outputs using minimal input data, significantly reducing the computational complexity. As future work, the framework has strong potential to be extended for generating longer video sequences, thereby improving the temporal resolution of simulation visualizations. Although this version was specifically trained on the water saturation property, future iterations could be generalized to accommodate a wider range of dynamic reservoir properties, including pressure, oil saturation, or temperature. Furthermore, we envision expanding the model beyond 2D interpolation by reconstructing multiple views to create full 3D visualizations of complex reservoir geometries. This opens the door to applications not just in visualization but also in predictive simulation, where the model could be conditioned on various physical parameters to forecast subsurface fluid behavior over time. Ultimately, this work lays the foundation for an efficient, scalable, and AI-augmented approach to reservoir simulation, offering engineers a powerful tool for understanding complex spatiotemporal dynamics with improved resolution, reduced costs, and greater interpretability.

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References

Chen, L., Cun, X., Li, X., He, X., Yuan, S., Chen, J., … Yuan, L. (2025). Sci-Fi: Symmetric Constraint for Frame Inbetweening. doi:https://doi.org/10.48550/arXiv.2505.21205

Croitoru, F.-A., Hondru, V., Ionescu, R. T., & Shah, M. (2023). Diffusion Models in Vision: A Survey. *IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE.* doi:https://doi.org/10.1109/TPAMI.2023.3261988

Doloi, N., Ghosh, S., & Phirani, J. (2023). Super-Resolution Reconstruction of Reservoir Saturation Map with Physical Constraints Using Generative Adversarial Network. SPE Reservoir Characterisation and Simulation Conference and Exhibition. Abu Dhabi. doi:https://doi.org/10.2118/212611-MS

Dong, J., Ota, K., & Dong, M. (2023). Video Frame Interpolation: A Comprehensive Survey. *ACM Transactions on Multimedia Computing, Communications and Applications*, 19. doi:https://doi.org/10.1145/3556544

Luan, T. (2024). A Comprehensive Review of Simulation Technology: Development, Methods, Applications, Challenges and Future Trends. *International Journal of Emerging Technologies and Advanced Applications.* doi:http://dx.doi.org/10.62677/IJETAA.2405119

Tulyakov, S., Gehrig, D., Georgoulis, S., Erbach, J., Gehrig, M., Li, Y., & Scaramuzza, D. (2021). Time Lens: Event-based Video Frame Interpolation. *Computer Vision and Pattern Recognition (CVPR)*, (pp. 16155–16164).

Unterthiner, T., Steenkiste, S.V., Kurach, K., Marinier, R., Michalski, M., & Gelly, S. (2019). FVD: A NEW METRIC FOR VIDEO GENERATION. The International Conference on Learning Representations (ICLR). Saudi Aramco: Public

Voleti, V., Jolicoeur-Martineau, A., & Pal, C. (2022). MCVD: Masked Conditional Video Diffusion for Prediction, Generation, and Interpolation. 36th Conference on Neural Information Processing Systems (NeurIPS 2022). doi:https://doi.org/10.48550/arXiv.2205.09853

Zhong, Z.Y., Sun, A., Wang, Y., & Ren, B. (2020). Predicting field production rates for waterflooding using a machine learning-based proxy model. *Journal of Petroleum Science and Engineering*, 194. doi:https://doi.org/10.1016/j.petrol.2020.107574.